

1) Propriedade 1: Soma das Integrais

↳ A integral de uma soma é igual à soma das integrais

$$\text{Ex: } \int (3x + 5x) dx = \int 3x dx + \int 5x dx$$

Propriedade 2: Regra da constante

↳ Quando você tem uma constante multiplicando a função, você pode remover ela da integral.

$$\text{Ex: } \int 3x dx = 3 \int x dx$$

$$\begin{aligned} 2) f'(x) &= 3x^2 - (\cancel{0} \cdot \cos(x^3) + \frac{3}{4}(-\sin(x^3) \cdot 3x^2)) + \frac{3x^2}{x^3} + e^{-4x} \cdot -4x^{-5} \\ f'(x) &= 3x^2 + \frac{9x^2}{4} \sin(x^3) + \frac{3x^2}{x^3} - e^{-4x} \cdot 4x^{-5} \end{aligned}$$

$$3-a) \int \left(x - \frac{1}{x}\right)^2 dx$$

$$\int \left(x - \frac{1}{x}\right) \left(x - \frac{1}{x}\right) dx$$

$$\int x^2 - 1 - 1 + \frac{1}{x^2} dx$$

$$\int x^2 dx - \int 2 dx + \int \frac{1}{x^2} dx$$

$$\frac{x^3}{3} - 2x - \frac{1}{x} + C$$

$$3-b) \int \left(3\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx = 3 \int x^{\frac{1}{2}} dx + \int x^{-\frac{1}{2}} dx$$

$$2x^{\frac{3}{2}} + 2\sqrt{x} + C$$

$$3-c) \int \left(1 + \frac{1}{x} \right)^{-3} \left(\frac{1}{x^2} \right) dx \quad u = 1 + \frac{1}{x}$$

$$-\int u^{-3} du$$

$$\frac{u^{-2}}{2} + C$$

$$\frac{du}{dx} = -\frac{1}{x^2}$$

$$-du = \frac{1}{x^2} dx$$

$$3-d) \int (\cos x - \sin x)^2 dx$$

$$\int (\cos x - \sin x)(\cos x - \sin x) dx$$

$$\int (\cos^2 x - \sin x \cos x - \sin x \cos x + \sin^2 x) dx$$

$$\int 1 - \sin(2x) dx \quad u = 2x$$

$$\int 1 dx - \int \sin(u) du \quad \frac{du}{dx} = 2$$

$$x + \cos(2x) + C \quad \frac{du}{2} = dx$$

$$3-e) \int \left(\frac{\ln(x)}{x} - 4 \frac{e^{\sqrt{x}}}{\sqrt{x}} \right) dx$$

$$\int \frac{\ln x}{x} dx - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx \quad \begin{array}{l} u = \ln x \quad v = \sqrt{x} \\ du = \frac{1}{x} dx \quad 2 dv = \frac{1}{\sqrt{x}} dx \end{array}$$

$$\int u du - 8 \int e^v dv$$

$$\frac{(\ln(x))^2}{2} - 8e^{\sqrt{x}} + C$$

$$3-f) \int \left(\sec(4x) - \frac{e^{\frac{1}{x}} + 8}{x^2} \right) dx //$$

$$\int \sec(4x) dx - \int \frac{e^{\frac{1}{x}}}{x^2} dx + \int \frac{8}{x^2} dx$$

$$\frac{1}{4} \int \sec(u) du - \int e^v dv + \int 8 dv$$

$$\frac{1}{4} \ln|\sec(4x) + \tan(4x)| - e^{\frac{1}{x}} + \frac{8}{x} + C$$

$$3-g) \int \left(\frac{\sin(\ln(x))}{x} - 4 \cos(2x) \right) dx \quad u = 2x \quad v = \ln(x)$$

$$\int \frac{\sin(\ln(x))}{x} dx - 4 \int \cos(2x) dx \quad \frac{du}{2} = dx \quad dv = \frac{1}{x} dx$$

$$\int \sin(v) dv - 2 \int \cos(u) du$$

$$-\cos(\ln(x)) - 2 \sin(2x) + C$$

$$3-h) \int \frac{3}{x \cdot (\ln 3x)^3} - \frac{x}{\cos^2(x^2)} dx \quad v = \ln(3x)$$

$$3 \int \frac{1}{x \cdot (\ln 3x)^3} dx - \int \frac{x}{\cos^2(x^2)} dx \quad dv = \frac{1}{x} dx$$

$$3 \int (\ln(3x))^{-3} \cdot \frac{1}{x} dx - \int x \cdot \sec^2(x^2) dx \quad u = x^2$$

$$3 \int v^{-3} dv - \frac{1}{2} \int \sec^2(u) du \quad \frac{du}{dx} = 2x$$

$$-\frac{3 (\ln(3x))^2}{2} - \frac{1}{2} \tan(x^2) + C$$

$$3-i) \int e^{3x} \cdot \sin(2x) dx \quad \int u \cdot dv = u \cdot v - \int v du$$

$$\int e^{3x} \cdot \sin(2x) dx = e^{3x} \cdot \left(-\frac{1}{2} \cos(2x)\right) + \frac{3}{2} \int \cos(2x) \cdot e^{3x} dx$$

$$dv = \sin(2x) \\ v = -\frac{1}{2} \cos(2x)$$

$$= \int e^{3x} \cdot \cos(2x) dx = e^{3x} \cdot \left(\frac{1}{2} \sin(2x)\right) - \frac{3}{2} \int \sin(2x) \cdot e^{3x} dx$$

$$u = e^{3x} \\ du = 3e^{3x}$$

$$\int e^{3x} \cdot \sin(2x) dx = e^{3x} \cdot \left(-\frac{1}{2} \cos(2x)\right) + \frac{3}{2} \left[\frac{e^{3x}}{2} \sin(2x) - \frac{3}{2} \int \sin(2x) \cdot e^{3x} dx \right]$$

$$\int e^{3x} \cdot \sin(2x) dx = -\frac{e^{3x}}{2} \cos(2x) + \frac{3e^{3x}}{4} \sin(2x) - \frac{9}{4} \int e^{3x} \cdot \sin(2x) dx$$

$$\frac{13}{4} \int e^{3x} \cdot \sin(2x) dx = -\frac{e^{3x}}{2} \cos(2x) + \frac{3e^{3x}}{4} \sin(2x)$$

$$\int e^{3x} \cdot \sin(2x) dx = \frac{-2e^{3x} \cos(2x) + 3e^{3x} \sin(2x)}{13} + C$$

$$3-J) \int \frac{\cos(\ln(x^2))}{x} - 4 \cot g(bx) dx \quad \begin{matrix} u = bx \\ \frac{du}{b} = dx \end{matrix}$$

$$\int \frac{\cos(\ln(x^2))}{x} dx - 4 \int \cot g(bx) dx \quad \begin{matrix} v = \ln(x^2) \\ \frac{dv}{dx} = \frac{2}{x} \end{matrix}$$

$$\frac{1}{2} \int \cos(v) dv - \frac{4}{b} \ln|\sin(bx)| + C \quad \frac{dv}{dx} = \frac{1}{x} dx$$

$$\frac{1}{2} \cdot \sin(\ln(x^2)) - \frac{4}{b} \ln|\sin(bx)| + C$$